Tarika Birch

Creative Thinker. Design Enthusiast. Software Engineer.

tarikabirch@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

University of the West Indies, Cave Hill Campus **September 2021 - May 2025**
BSc. Computer Science with Management - First Class Honours
GPA: 3.89

WORK EXPERIENCE

GovTech Barbados - Hybrid, Barbados
Frontend Developer **February 2026 - Present**

- **Built Design Adapter System:** Designed and built a design adapter for the government forms renderer, letting different agencies create new forms with their own design system without building each form from scratch.
- **Automated Form Filling:** Built an automated form-filling tool that cut per-page completion time from 3–5 minutes to 10–30 seconds, speeding up testing and stakeholder demos by up to 97%.
- **Raised Test Coverage to ~95%:** Raised automated test coverage of the forms and backend services to about 95% and set minimum coverage thresholds in CI, fixing hidden bugs found along the way.

Supreme Court of Barbados - Remote, Barbados
Judiciary Software Engineer **September 2025 - February 2026**

- **Implemented RBAC Service:** Designed and implemented a centralized Role-Based Access Control service for the Court Management System, eliminating duplicated permission checks and reducing permission update time by ~90%.
- **Automated Jury Selection:** Designed and built a program that automates jury duty selection for the Supreme Court, replacing the manual selection process.
- **Developed Secure RESTful APIs:** Designed and implemented secure RESTful APIs using NestJS to support case management, ensuring reliable and maintainable backend services.

FluoretiQ - Bristol, UK
Machine Learning Fellow **July 2025 - August 2025**

- **Conducted Exploratory Data Analysis:** Performed exploratory data analysis on microscopy datasets to identify key features for bacterial classification.
- **Explored & Fine-Tuned Models:** Designed, trained, and evaluated machine learning models with Python, using performance metrics to guide iterative improvements.
- **Automated Data Preparation:** Automated the gathering, sorting and filtering of bacterial sample data to speed up model training.

District 6 Technologies - Remote, Barbados
Software Engineer **September 2024 - May 2025**

- **Prototyping & Wireframing:** Developed interactive prototypes with Figma to visualize design concepts and validate usability early in the product lifecycle.
- **Responsive Web Design:** Implemented responsive, mobile-first interfaces using HTMX and Tailwind CSS.
- **Requirements Gathering:** Engaged with clients to gather business requirements and translate them into intuitive solutions.

Pellucid IT - Remote, Barbados
Software Developer **June 2024 - January 2025**

- **Enhanced UI/UX:** Designed and developed the Pellucid Website using Drupal.
- **Established CI/CD Pipelines:** Built a custom deployment module for the Pellucid Website, automating configuration management and streamlining CI/CD workflows.
- **Professional Mentorship:** Mentored a Software Development Intern, conducting code reviews, ensuring adherence to best practices and contributing to their professional growth.

Software Development Intern **January 2024 - June 2024**

- **Improved Operational Efficiency:** Designed and developed a modern and intuitive user interface for the CPAG Biodiversity Dashboard with React, streamlining Protected Areas management and optimizing user workflows.
- **Streamlined Development Process:** Assisted in merging the CPAG Biodiversity and Conservation Dashboard codebases, removing duplicated modules and improving maintainability.
- **Ensured Code Quality and Consistency:** Actively participated in code reviews to ensure adherence to high-quality standards and maintain consistency across all CPAG platforms.

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SKILLS & INTERESTS

- **Programming Languages** - JavaScript, TypeScript, Python, HTML, CSS, SQL
- **Frameworks** - React, React Native, Angular, Flutter, NestJS, Node.js, Tailwind CSS
- **Databases** - PostgreSQL, Firestore
- **Tools & Cloud Services** - Git, Docker, Firebase
- **Design Skills** - Figma, UI/UX, Prototyping, Wireframing
- **Soft Skills** - Critical Thinking, Problem-Solving, Time Management, Attention to Detail

PROJECTS

Flexa **Professional Project**

Flutter | *Electron* | *Firebase* October 2025 - Present

- Built a cross-platform stretch reminder app using Flutter and Electron, including background scheduling logic and highly customizable reminder workflows.
- Implemented multi-device synchronization using Firebase to ensure real-time updates across desktop and mobile clients.

Tessera Web **Professional Project**

React | *TypeScript* | *Tailwind* | *Vite* October 2025

- Designed and developed a production-ready company website using React and TypeScript, delivering a fast, responsive, and component-driven user experience.
- Implemented a mobile-first UI with Tailwind CSS, ensuring consistency and accessibility across screen sizes.
- Deployed and maintained the live website on Netlify, ensuring stable and reliable access for end users.

Re-Source **Personal Project**

Flutter | *Firebase* | *Gemini AI* July 2025 - November 2025

- Designed and developed a cross-platform Flutter app that organizes and retrieves useful links, using Gemini AI to categorize saved links and parsing link metadata automatically.
- Implemented Firebase authentication with real-time sync across devices.
- Released Re-Source on the Google Play Store after extensive internal, closed, and public testing, ensuring a reliable and user-friendly launch.

Eye Disease Classifier **Personal Project**

Python | *Streamlit* June 2025

- Developed an Eye Disease Classifier using machine learning, achieving 90% accuracy in detection and classification.
- Implemented a Streamlit-based web interface enabling real-time image uploads and live classification results.
- Created setup and onboarding documentation, including installation steps, environment configuration, and project structure guidelines to help users clone, run, and train the model easily.

AWARDS & ACHIEVEMENTS

- 2025 Winner of the Windrush I Accelerator Program - Microsoft UK and UWI **June 2025**
- 3rd Place - Barbados AI Fraud Detection Hackathon - Zindi **May 2025**
- Faculty of Science and Technology Student Award - FST, UWI **October 2022, October 2023**
- Incentive Award for Excellence In Visual Arts - National Independence Festival of Creative Arts **November 2019**

LEADERSHIP

- Vice President - Computer Science Society, Faculty of Science and Technology, UWI **January 2025 - May 2025**
- Secretary - Computer Science Society, Faculty of Science and Technology, UWI **September 2023 - January 2025**

CERTIFICATES

- UX Design Fundamentals - Coursera
- Visual Elements of User Interface Design - Coursera
- JavaScript Basics - Coursera
- TypeScript - Coursera
- Frontend Development using Angular - Coursera
- The Nest JS Bootcamp: Complete Developer Guide - Udemy